

Hardware qualification plan: head-roll axis, frozen baseline 7d857df507c389c9

Status: Proposed, not performed. Limits below are provisional project values, to be confirmed against the real application and fixture. Passing qualifies the tested envelope only; the adaptive correction is not hardware-tested unless it first passes a new registered simulation comparison.

Instrumentation (one timebase, ≥ 1 kHz): roll and yaw encoders; current target, pre-clamp command, measured current and active limit; bus voltage, phase voltage/PWM duty and temperature; CAN send/receive timestamps; drive mode and fault state; original and governed references. Fixture: calibrated torque sensor, hard-wired e-stop independent of the host, supported load, verified clearance, and mechanical stops or a verified brake.

Global abort triggers: stale/lost feedback timestamps; wrong torque sign or unexpected current/voltage behavior; tracking error $> 5^\circ$; $\geq 95\%$ current utilization for > 100 ms; unexplained latched fault; fixture temperature limit (below the simulated 130°C trip); motion outside verified clearance. End the test and isolate actuation.

Communication loss gives **drive-local damping only**, which does not hold against gravity; there is **no promised position hold** after feedback loss or e-stop, so the support, brake or stops must contain motion (simulated loaded outage: up to 42° , a warning, not a bound).

#	Test (in order; each gates the next)	Stage-specific stops	Acceptance to proceed
1	Blocked-axis identification: torque sign, current offset, K_t from torque/current, stationary electrical checks and command/feedback latency. Speed is zero, so this step cannot identify K_e .	Wrong sign or unexplained current/voltage; K_t outside $\pm 15\%$ model range; > 4 ms latency bursts	Confirm sign, K_t , sensing and protected stop path before any rotation; otherwise update the model and repeat the Task 2 analysis.
2	Guarded low-speed rotation , supported axis, initially within $\pm 5^\circ$ at the derated limit: measure K_e primarily as open-circuit EMF while the unpowered axis is back-driven (within $\pm 5^\circ$ the axis stays slow, so $K_e \cdot \dot{q}$ is small next to the current-dependent IR drop and R is uncertain by $\pm 25\%$), cross-check it from voltage/current/speed, and resolve the bus-to-phase/PWM convention.	Unexpected motion or voltage; margin to clearance lost	K_e and usable voltage at speed measured; re-run the electrical and delay limits if they differ from the model before broader moves.
3	Static holds , supported axis at 2.4 A: nominal then identified loads, 0° and $\pm 15^\circ$.	Holding torque ($K_t \cdot \text{measured current}$) above $0.8 \cdot K_t I_{\text{lim}} - 0.05$ N·m	Hold error $\leq 0.5^\circ$. Screen loads using measured holding demand; the nominal governor cannot infer an unknown payload.
4	Slow moves , $\pm 15^\circ \rightarrow \pm 45^\circ$, nominal load, then 3.2 A; then a small 3 Hz sine at nominal and added command delay, timestamped (Task 5).	Error $> 5^\circ$; sustained limit	$\leq 2^\circ$ RMS and $\leq 5^\circ$ peak against the governed reference; $\geq 95\%$ path progress; no faults; report original-request error too. Task 5: report measured delay and 3 Hz gain/phase change against its prediction.
5	Yaw disturbance , roll held: low-amplitude sweep, then B/C frequencies only if the fitted coupling predicts $\geq 20\%$ current reserve.	Coupling above reserve	Validate the fit at a held-out frequency (predicted residual within $\pm 20\%$); $\leq 2^\circ$ RMS in the admitted envelope; record any yaw reduction.
6	Faults and derating on the protected fixture: command-only, feedback-only and both-direction outages; $3.2 \rightarrow 2.4$ A mid-move; tracking-fault catch.	Excursion approaches verified clearance	No target-current limit bypass; 50 ms fresh aligned dwell before re-arm; faulted request stays suspended; measured passive fallback travel $\leq 50\%$ of the verified clearance in every outage; re-arm within 0.2 s of fresh commands.
7	Payload and envelope: D-like shifted load at reduced speed, then loaded finite move.	As above	Completed requests: $\leq 2^\circ$ RMS / $\leq 5^\circ$ peak. Reshaped requests: governed-reference peak $\leq 5^\circ$ and the original-request error reported. Rejected requests: reason logged. No current-limit bypass; no unexplained fault.